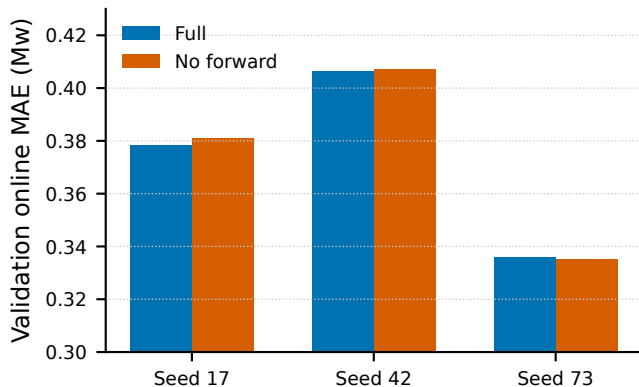


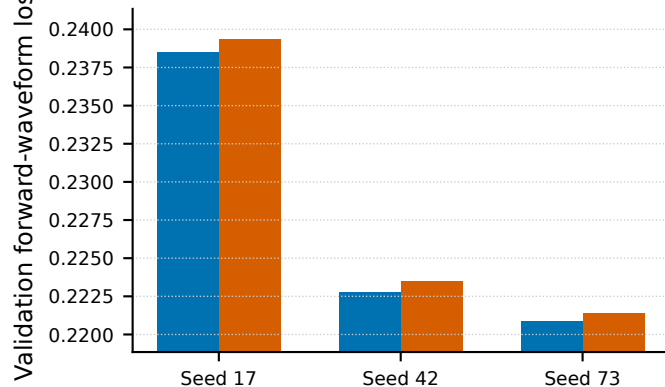
Matched no-forward-loss ablation

A

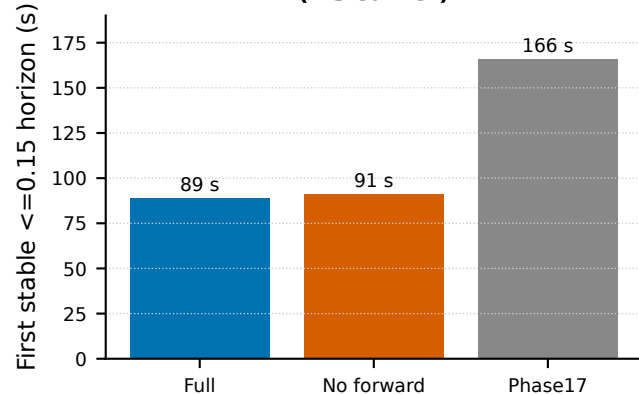
**Validation online MAE
(seed-dependent effect)**

**B**

**Forward-waveform mismatch
(lower with full loss)**

**C**

**Stable ≤ 0.15 horizon
(2 s earlier)**

**D**

**External online gain
(millimagnitude scale)**

